

DAY 7 HDOC
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BATCH - 15

Question 1

Aim

To write a C program to find the **relative velocity in the same direction** when two velocities are given.

Step 1: Problem Understanding

Input:

- Velocity of first object (V1)
- Velocity of second object (V2)

Output:

- Relative Velocity

Formula

Relative Velocity = $|V1 - V2|$

Step 2: Test Cases

Test Case	Input(s)	Expected Output	Actual Output	Result
1	V1=60, V2=40	20	20	Pass
2	V1=100, V2=80	20	20	Pass
3	V1=45, V2=70	25	25	Pass

Step 3: Algorithm

1. Start.
2. Declare variables V1, V2 and RV.
3. Read V1 and V2.
4. If $V1 > V2$, calculate $RV = V1 - V2$.
5. Otherwise calculate $RV = V2 - V1$.
6. Display the relative velocity.
7. Stop.

Step 4: C Program

```
#include<stdio.h>

int main()

{

    float V1, V2, RV;

    printf("Enter first velocity: ");

    scanf("%f",&V1);

    printf("Enter second velocity: ");

    scanf("%f",&V2);

    if(V1 > V2)

        RV = V1 - V2;

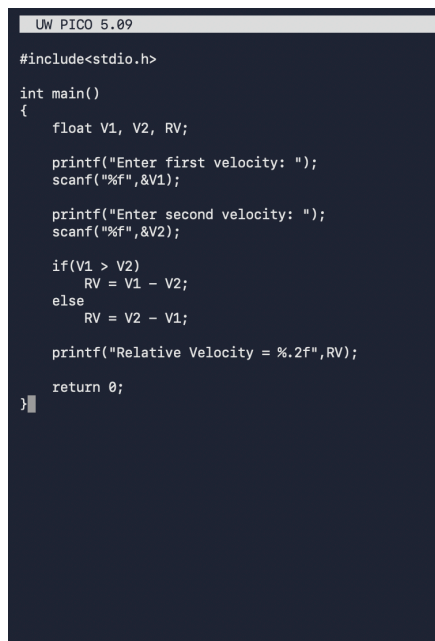
    else

        RV = V2 - V1;

    printf("Relative Velocity = %.2f",RV);

    return 0;

}
```

A screenshot of a code editor window titled "UW PICO 5.09". The editor contains the same C program as shown in the previous block. The code is as follows:

```
#include<stdio.h>

int main()
{
    float V1, V2, RV;

    printf("Enter first velocity: ");
    scanf("%f",&V1);

    printf("Enter second velocity: ");
    scanf("%f",&V2);

    if(V1 > V2)
        RV = V1 - V2;
    else
        RV = V2 - V1;

    printf("Relative Velocity = %.2f",RV);

    return 0;
}
```

Step 5: Compilation and Execution

Compilation Status: Successful

```
[bhavya@Bhavyas-MacBook-Air ~ % nano 7q1.c
[bhavya@Bhavyas-MacBook-Air ~ % clang 7q1.c -o 7q1.c
[bhavya@Bhavyas-MacBook-Air ~ % ./7q1.c
Enter first velocity: 60
Enter second velocity: 40
Relative Velocity = 20.00%
bhavya@Bhavyas-MacBook-Air ~ %
```

Question 2

Aim

To write a C program to calculate **Weight** when mass and acceleration due to gravity are given.

Step 1: Problem Understanding

Input

- Mass (m)
- Acceleration due to gravity (g)

Output

- Weight (W)

Formula

Weight = Mass $\times$ Gravity

$$\begin{aligned} m &= 8\text{kg} \\ F_g &= 78.4\text{N} \\ g &= 9.8 \text{ m/s}^2 \\ F_g &= m \cdot g \end{aligned}$$

Step 2: Test Cases

Test Case	Input(s)	Expected Output	Actual Output	Result
1	m=10, g=9.8	98	98	Pass
2	m=5, g=9.8	49	49	Pass
3	m=20, g=9.8	196	196	Pass

Step 3: Algorithm

1. Start.
2. Declare variables mass, gravity and weight.
3. Read mass and gravity.
4. Calculate weight = mass × gravity.
5. Display weight.
6. Stop.

Step 4: C Program

```
#include<stdio.h>

int main()

{

    float mass, gravity, weight;

    printf("Enter mass: ");

    scanf("%f",&mass);

    printf("Enter gravity: "); scanf("%f",&gravity);

    weight = mass * gravity;

    printf("Weight = %.2f",weight);

    return 0;

}
```

```
UW PICO 5.09 File: 7q2.c
#include<stdio.h>

int main()
{
    float mass, gravity, weight;

    printf("Enter mass: ");
    scanf("%f",&mass);

    printf("Enter gravity: ");
    scanf("%f",&gravity);

    weight = mass * gravity;

    printf("Weight = %.2f",weight);

    return 0;
}
```

Step 5:Compilation and Execution

Compilation Status: Successful

```
[bhavya@Bhavyas-MacBook-Air ~ % nano 7q2.c
[bhavya@Bhavyas-MacBook-Air ~ % nano 7q2.c
[bhavya@Bhavyas-MacBook-Air ~ % clang 7q2.c -o 7q2.c
[bhavya@Bhavyas-MacBook-Air ~ % ./7q2.c
Enter mass: 10
Enter gravity: 9.8
Weight = 98.00%
bhavya@Bhavyas-MacBook-Air ~ %
```

Question 3

Aim

To write a C program to calculate **Electrical Energy** when power and time are given.

Step 1: Problem Understanding

Input

- Power (P)
- Time (T)

Output

- Electrical Energy (E)

Formula

Energy = Power $\times$ Time

Step 2: Test Cases

Test Case	Input(s)	Expected Output	Actual Output	Result
1	P=100, T=2	200	200	Pass
2	P=250, T=4	1000	1000	Pass
3	P=500, T=5	2500	2500	Pass

Step 3: Algorithm

1. Start.
2. Declare variables power, time and energy.
3. Read power and time.
4. Calculate energy = power $\times$ time.
5. Display energy.
6. Stop.

Step 4: C Program

```
#include<stdio.h>

int main()

{

float power, time, energy;

printf("Enter power: ");

scanf("%f",&power);

printf("Enter time: ");

scanf("%f",&time);

energy = power * time;

printf("Electrical Energy = %.2f",energy);

return 0;

}
```

```
UW PICO 5.09

#include<stdio.h>

int main()
{
    float power, time, energy;

    printf("Enter power: ");
    scanf("%f",&power);

    printf("Enter time: ");
    scanf("%f",&time);

    energy = power * time;

    printf("Electrical Energy = %.2f",energy);

    return 0;
}
```

STEP 5 : Compile and execute the program, then evaluate the code against the test case

Compilation and Execution

Compilation Status: Successful

```
[bhavya@Bhavyas-MacBook-Air ~ % nano 7q3.c
[bhavya@Bhavyas-MacBook-Air ~ % clang 7q3.c -o 7q3.c
[bhavya@Bhavyas-MacBook-Air ~ % ./7q3.c
Enter power: 100
Enter time: 2
Electrical Energy = 200.00%
[bhavya@Bhavyas-MacBook-Air ~ % ]
```

